

[I225] Statistical Signal Processing(E) Office Hour 6

1. Suppose a zero-mean wide-sense stationary random process $S(t)$ is predicted by the following linear predictor:

$$\hat{S}(t + \tau) = aS(t).$$

Assume that the autocorrelation values are

$$R_{SS}(0) = 10,$$

and

$$R_{SS}(\tau) = 6.$$

- (a) Write the prediction error.
 - (b) Use the orthogonality principle to derive the optimal coefficient a .
 - (c) Compute the corresponding minimum mean square error (MSE).
 - (d) Explain how the prediction performance changes when $R_{SS}(\tau)$ becomes smaller.
2. A zero-mean wide-sense stationary signal $S(t)$ has the following power spectral density:

$$S_{SS}(\omega) = \frac{4}{\omega^2 + 4}.$$

The signal is corrupted by additive white noise $v(t)$ whose power spectral density is

$$S_{vv}(\omega) = 1.$$

Assume that the signal and the noise are uncorrelated. The observed process is

$$X(t) = S(t) + v(t).$$

- (a) Derive the optimal non-causal Wiener filter $H(\omega)$.
- (b) Compute the value of $H(0)$.
- (c) Compute the minimum MSE of the Wiener filter.
- (d) Explain why high-frequency components are more strongly suppressed.

3. Consider the following one-step linear predictor:

$$\hat{X}[2] = -a_0X[0] - a_1X[1].$$

The autocorrelation sequence of the wide-sense stationary signal is given by

$$r[0] = 4, \quad r[1] = 2, \quad r[2] = 1.$$

- (a) Write the prediction error.

(b) Apply the orthogonality principle to derive the Yule-Walker equations.

(c) Express the equations in matrix form.

(d) Solve for a_0 and a_1 .

(e) Explain why the coefficient matrix is called a Toeplitz matrix.

4. Consider the continuous-time signal

$$x(t) = \cos(2 \cdot \pi \cdot 400 \cdot t).$$

(a) Determine the frequency f in Hz.

(b) Determine the angular frequency ω in rad/s.

(c) Write the Fourier transform $X(f)$.

(d) Find the minimum sampling frequency required by the Nyquist-Shannon sampling theorem.

(e) Determine whether sampling at 600 Hz can perfectly reconstruct the signal.

5. Explain the concept of aliasing. We want to estimate the parameter vector

$$\boldsymbol{\theta} = \begin{bmatrix} \theta_1 \\ \theta_2 \end{bmatrix}.$$

The prior estimate is

$$\hat{\boldsymbol{\theta}}_0 = \begin{bmatrix} 0 \\ 0 \end{bmatrix},$$

and the prior covariance matrix is

$$M_0 = \begin{bmatrix} 2 & 0 \\ 0 & 2 \end{bmatrix}.$$

A scalar measurement is obtained through

$$x_1 = \mathbf{h}^T \boldsymbol{\theta} + w_1,$$

where

$$\mathbf{h} = \begin{bmatrix} 1 \\ 1 \end{bmatrix},$$

and the measurement noise variance is

$$\sigma_w^2 = 1.$$

The observed value is

$$x_1 = 3.$$

(a) Compute the Kalman gain vector K_1 .

- (b) Compute the updated estimate vector.
- (c) Compute the updated covariance matrix.
- (d) Explain the role of the Kalman gain in the update process.